Recap of the Python part

Programming & Data Management, Part II (Python)

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What the exam asks, and how to practise it

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- The exam asks you to read code and predict what it does, without a computer. That is what we practise today.

The four sessions

Session 1: a project someone else can rerun

- Project layout: raw data in `data/raw/`, never edited; code in `scripts/`; paths built from the project folder (`ROOT / "data" / "raw"`), so the project runs on any computer
- Python basics: `"20" + "24"` joins text; lists count from 0; a slice `x[1:3]` stops *before* position 3
- Objects, not commands: `df.head()` asks the object to act; `df.sort_values("x")` changes nothing until you write `df = df.sort_values("x")`

- In pandas, nothing changes unless you assign (`replace`).

Session 2: look at the data before you use it

- The intake checklist: type, missing values, range. One text value such as `"****"` makes the whole column text
- Sentinel codes (999, -9) are numbers, so `isna()` cannot see them; set them to missing with `.mask()` or `.where()`
- Aggregate: `groupby().agg()` gives one row per group; `groupby().transform()` keeps every row and adds the group value

- The errors that matter most give no error message: check type, missing and range every time.

Session 3: combine tables without losing rows

- Keys: unique and never missing; the same type in both tables (`1` and `"1"` do not match)
- Merge audit: `how="left"`, `validate="m:1"`, `indicator=True`; predict the number of rows before you run
- Wide and long: `melt` gives one row per unit and column melted; years taken from column names are text. A lag is `groupby("pid")["x"].shift(1)`

- Before every merge: is the key unique, and how many rows do you expect?

Session 4: get data, and rebuild it all

- API: save the raw answer first; check that every row arrived (`raise_for_status()`, a count against the metadata)
- Dates: `to_datetime`, `resample("QE")` with a stated rule (mean for prices, sum for flows), drop incomplete quarters; year-on-year growth is `pct_change(4)` for quarters
- Scale up: functions, loops over files, `run_all.py`, the regression and DuckDB are how you work, but not on this year's exam

- Save the raw answer first, and check that every row arrived.

The exam

What the exam looks like

- One exam for both parts: WISEflow, in an exam hall, 2 hours, no aids; 100 points, 50 to pass
- The Python part: 16 questions, 40 points; some give 2 points, some 3, and the paper shows the points at each question
- Not on the exam this year: your own functions, loops over files, `run_all.py`, the regression, DuckDB
- One correct answer per question; a wrong answer costs nothing, so answer them all
- Most questions show a few lines of code; a few are on the readings
- The Blackboard practice exam is from an older course: use the sample exam on the course site

- No computer: the skill is to read the code and predict.

Three questions

Predict on paper: how many rows does `df` have?

Alone, one minute; then hands up for A–D

```
df = pd.DataFrame({"region": ["N", "N", "S"],
                   "wage": [100, 200, 90]})
df["mean_wage"] = (df.groupby("region")["wage"]
                     .transform("mean"))
```

A. 2 B. 6 C. 3 D. 1

Predict on paper: how many rows does `df` have?

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```

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Answer: C, 3 rows. `transform` keeps every row and gives each the mean of its region: 150, 150, 90 (Stata: `bysort region: egen`). `agg` would collapse to 2 rows.

Predict on paper: how many rows does `long` have?

Alone, one minute; then hands up for A–D

```
wide = pd.DataFrame({"pid": [1, 2, 3],
                     "wage2023": [100, 110, 90],
                     "wage2024": [105, 112, 95]})
long = wide.melt(id_vars="pid", var_name="year",
                 value_name="wage")
```

A. 3 B. 2 C. 9 D. 6

Predict on paper: how many rows does `long` have?

Alone, one minute; then hands up for A–D

```
wide = pd.DataFrame({"pid": [1, 2, 3],
                     "wage2023": [100, 110, 90],
                     "wage2024": [105, 112, 95]})
long = wide.melt(id_vars="pid", var_name="year",
                 value_name="wage")
```

A. 3 B. 2 C. 9 D. 6

Answer: D, 6 rows. One row per person and melted column: $3 \times 2 = 6$ (Stata: `reshape long`). And `year` holds the text `"wage2023"`: a number only after you clean and convert it. The reverse, long to wide, is `pivot` (Stata: `reshape wide`).

Predict on paper: what does the last line print?

Alone, one minute; then hands up for A–D

```
origin = pd.Series([1, 3, 2])
names = {1: "US", 2: "Europe", 3: "Japan"}
region = origin.map(names)
print(region.iloc[1])
```

A. Europe B. 3 C. US D. Japan

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A. Europe B. 3 C. US D. Japan

Answer: D, Japan. `map` looks up each code in the dictionary; `iloc[1]` is the *second* row (Python counts from 0), code 3.

How to prepare

1. Do the sample exam on the course site in 45 minutes, on paper, then check
2. Watch the six videos (course site, *Videos*) and answer the questions under each before you open the answers
3. Do each session's self-test slide without AI
4. Redo the labs from scratch, without AI: predict each output before you run
5. Revisit the readings: the course site lists five required articles per session. An exam question or two may touch them, and they explain the *why* behind the code.

– Predict, then check. That is the exam, and it is how the course works.